



Research paper

Evaluation of multiple immunoassay formats for detection of anti-drug antibodies to zinpentraxin alfa

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ABSTRACT

Zinpentraxin alfa (rhPTX-2; PRM-151) is currently being developed for the treatment of fibrotic diseases such as idiopathic pulmonary fibrosis and myelofibrosis. Notably, because it is administered chronically and has an endogenously expressed counterpart, clinical studies of zinpentraxin alpha must include immunogenicity assessments. Since the typical homogenous bridging ELISA assay does not adequately measure anti-drug antibodies (ADAs) against zinpentraxin alfa, additional assay formats have been developed to evaluate immunogenicity of this therapeutic. Here, we present the evaluation of four distinct assay formats that were used to measure zinpentraxin alpha ADA: step-wise bridging, direct binding, total ADA, and the semi-homogeneous formats, based on multiple parameters including assay sensitivity, precision, and drug tolerance. This paper presents the full details of method development for each of the aforementioned assay formats including evaluation of sample pre-treatment, determination of cut point, and assessment of assay performance by analyzing a subset of clinical samples. Overall, the semi-homogenous ADA assay format with no sample pre-treatment was selected for the measurement of zinpentraxin alpha immunogenicity as it provided the desired sensitivity, drug tolerance, and reproducibility. Our study emphasizes the importance of assay format evaluation during drug development and the necessity to select the most suitable assay format and sample pre-treatment method by which to evaluate therapeutic drug immunogenicity.

1. Introduction

Zinpentraxin alfa is a recombinant form of a native protein called pentraxin-2 (PTX-2), also known as recombinant human serum amyloid P (Raghu et al., 2022). Zinpentraxin alfa is being investigated as a potential treatment for fibrotic diseases such as idiopathic pulmonary fibrosis (IPF) and myelofibrosis (MF) (Duffield and Lupher Jr., 2010; Dillingh et al., 2013; Verstovsek et al., 2018; Raghu et al., 2019), and was granted FDA breakthrough therapy designation based on the data from a double-blind, multicenter phase 2 clinical trial that compared zinpentraxin alfa treatment with placebo in 111 IPF patients (Raghu et al., 2018). Like the native protein (Lupher Jr., 2012), zinpentraxin alfa is expressed and purified as a non-covalent, homo-pentameric glycoprotein. In native PTX-2, each glycan exhibits a typical complex

biantennary structure with α 2,6-linked bi-sialylated species being the predominant form. Zinpentraxin alfa also exhibits a typical complex biantennary structure as the predominant glycoform similar to the native protein, while additionally exhibiting α 2,3 sialic acid linkages, which are typical for recombinant proteins that are expressed in CHO cell lines. Because the endogenous PTX-2 plays a significant role in modulating fibrosis, inflammation, and proliferation (MacDonald and Kilpatrick, 2006; Pilling et al., 2007; Castano et al., 2009), the consequences of neutralizing antibodies induced by zinpentraxin alfa treatment could be severe if ADAs cross-react to the endogenous counterpart.

Immunogenicity monitoring is essential in clinical trials as immunogenic responses to therapeutics have been found to result in altered drug efficacy and to trigger unwanted safety events that can differ among patients (Koren et al., 2002; Schellekens, 2002; Schellekens and

Abbreviations: ACE, affinity capture elution; ADAs, anti-drug antibodies; CDR, complementarity-determining regions; CP, cut point; CPF, cut point factor; ECL, electrochemiluminescent; HRP, horseradish peroxidase; IPF, idiopathic pulmonary fibrosis; mAb, monoclonal antibody; MF, myelofibrosis; NC, negative control; OD, optical density; pAb, polyclonal antibody; Panda, precipitation and acid dissociation; PBS, phosphate-buffered saline; PC, positive control; PMF, primary myelofibrosis; PTX-2, pentraxin-2; RT, room temperature; SA, streptavidin; SD, standard deviation; SPEAD, solid phase with extraction acid dissociation; TMB, tetramethylbenzidine.

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Casadevall, 2004; Gunn 3rd et al., 2016). Therefore, immunogenicity analysis is mandated by regulatory agencies throughout the drug development, clinical studies, and post-launch phases (Agency, 2015; 2019). A variety of different assay formats have been developed and validated for ADA detection which include direct binding assays (Mire-Sluis et al., 2004), bridging immunoassays (Patton et al., 2005), affinity capture elution (ACE) assays (Bourdage et al., 2007), and precipitation and acid dissociation (Panda) assays (Zoghbi et al., 2015). Among these, the bridging immunoassay is the most commonly conducted format in which drugs conjugated with different tags are used as capture and detection reagents to form a sandwich with the ADAs. Advantages of the bridging assay format include good assay sensitivity, ability to detect a majority of ADA immunoglobulin isotypes (Aalberse et al., 1999), and operational simplicity for high throughput analysis (Hafeez et al., 2018; Singh et al., 2018; Lu et al., 2020). A disadvantage of the bridging assay format is its vulnerability to interference from therapeutic drugs, which can bind to ADAs and hinder the formation of a complex with the capture and detection reagents, potentially leading to false negative results. In addition, drug targets can also interfere, leading to either false positive results when multimeric soluble targets are present, or false negative results in the presence of monomeric soluble targets.

A direct electrochemiluminescent (ECL) assay using solid phase with extraction acid dissociation (SPEAD) sample treatment (Patton et al., 2005; Smith et al., 2007) was developed by Promedior (part of Roche, Basel, Switzerland) for the detection of anti-zincpentraxin alfa antibodies in human serum and implemented in a phase 2 study (Tremblay and Mascarenhas, 2021; Verstovsek et al., 2023) in subjects with primary

myelofibrosis (PMF). To allow ADA binding, this assay requires overnight incubation of sample and biotinylated zincpentraxin alfa (Butterfield et al., 2010). The immune complexes are then captured on a streptavidin plate, followed by washing and acid treatment to break apart the ADA binding. Dissociated ADAs are transferred to a second plate for detection using Sulfo-tagged zincpentraxin alfa in a direct ECL assay. While this assay format exhibits good drug tolerance, the SPEAD treatment requires significant manipulation of the samples, which is time-consuming and potentially increases the assay variability and loss of low binding antibodies. In this regard, greater variability was observed in positive controls. Interference from proteins bound to the drug is another potential limitation of the SPEAD assay method due to the fact that this format does not confirm the existence of immunoglobulins directed to the drug, which could lead to a false positive signal. Considering the aforementioned limitations of the SPEAD method, the first format evaluated was the most commonly used homogeneous bridging immunoassay. However, this format had high background signal which led to poor assay sensitivity. This led us to evaluate alternative assay formats that could provide dependable ADA detection while meeting the analytical performance criteria laid out in regulatory guidance documents. Fig. 1 depicts a diagrammatic representation of each assay format.

The step-wise bridging format (Fig. 1a) is similar to the homogeneous bridging assay format with ADA and drug binding taking place on a solid surface. In this format, drug is coated on the plate, followed by sample incubation. Captured ADAs are detected using biotinylated drug.

The direct binding format (Fig. 1b) employs a similar protocol as the

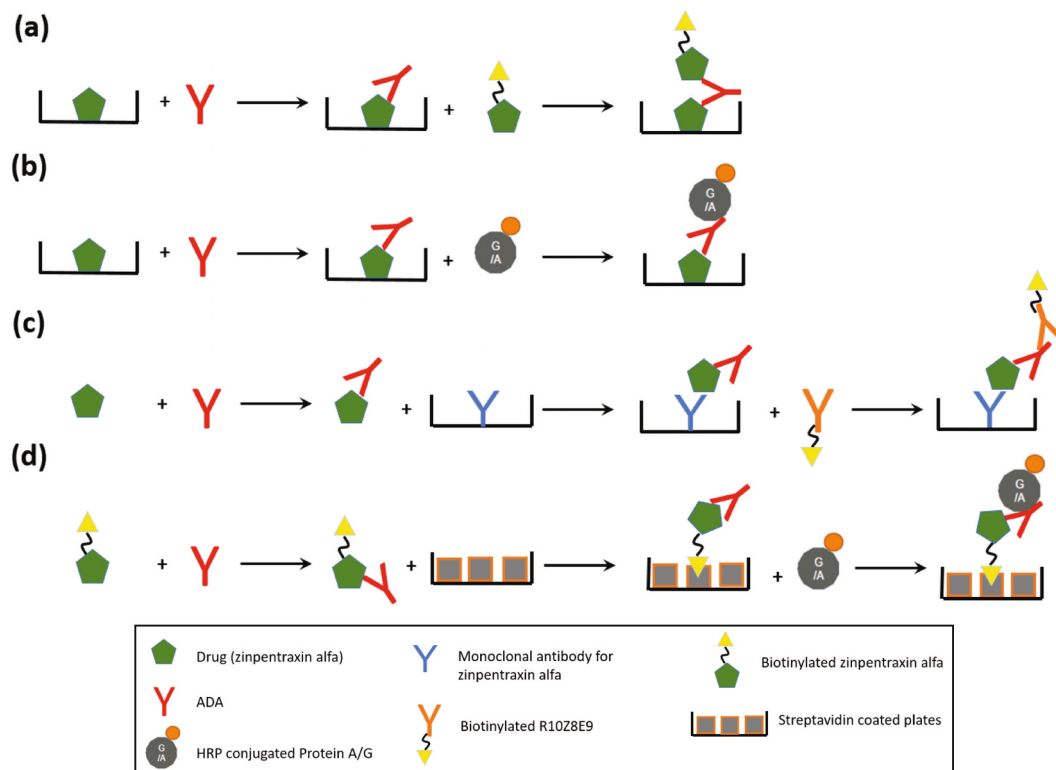


Fig. 1. ADA assay formats for detection ADAs against zincpentraxin alfa. (a) Step-wise bridging format. Drug is coated on the plate. Test sample is added, and ADAs are captured by plate-bound drug. After washing to remove any unbound reagents and sample, bound ADAs are detected using biotinylated drug, signal is revealed by SA-HRP followed with TMB. (b) Direct binding format. Drug is coated on the plate. Test sample is added, and ADAs are captured by plate-bound drug. After washing to remove any unbound reagents and sample, bound ADAs are identified by HRP conjugated Protein A/G, signal is detected with TMB. (c) Total ADA format. This assay format involves the addition of drug to the samples. The Drug/ADA complexes are captured using a monoclonal antibody against the drug pre-coated on the plate. ADAs are detected using a biotinylated antibody that can bind to different subclasses of human IgG. Signal is detected by SA-HRP followed with TMB. (d) Semi-homogeneous format. The sample is mixed with the biotinylated drug. Streptavidin-coated plates are then used to capture the biotin-drug/ADA complexes. ADAs are detected using HRP conjugated Protein A/G, and signal is detected using TMB. SA-HRP = Streptavidin-horseradish peroxidase conjugate; TMB = 3,3,5,5'-Tetramethylbenzidine.

step-wise bridging format while using Protein A/G as the detection reagent, which binds IgG, IgA, and IgM with different affinities (Intaramat et al., 2016). In this assay format, an assay blocker such as ChonBlock is used at both plate blocking and sample dilution steps to reduce the non-specific background signal.

The total ADA format (Fig. 1c) employs the addition of drug to bind and capture ADAs before transferring to an ELISA plate pre-coated with a monoclonal antibody against the drug. This method makes it possible to detect both free and bound ADAs in samples regardless of the level of drug present in the sample. In this format, anti-human IgG monoclonal antibody (Clone R10Z8E9, Genentech) is used as the detection reagent.

The semi-homogenous format (Fig. 1d) employs incubating samples overnight with biotinylated drug to allow the binding of ADAs. The immune complexes are then captured on a streptavidin plate followed by ADA detection using HRP-conjugated Protein A/G.

In this paper, we present results from evaluation of the sensitivity, precision, and drug tolerance of the four ADA assay formats (Fig. 1). A subset of samples from a phase 2 study of zinpentraxin alfa in subjects with primary myelofibrosis was selected and analyzed using different assay formats to further evaluate and compare their performance. The objective of this study was to assess the critical parameters of each format to identify the one most suitable for subsequent validation and analysis of clinical samples in the future.

2. Materials and methods

2.1. Reagents and serum samples

Zinpentraxin alfa, a recombinant form of native pentraxin-2, was generated at ACG Biologics (Bothell, WA, USA). Zinpentraxin alfa was biotinylated at Genentech (South San Francisco, CA, USA). The capture reagent used in the total ADA assay format was a rabbit IgG monoclonal antibody against zinpentraxin alfa generated at Genentech. The detection reagent for the total ADA assay format was a biotinylated murine IgG1 monoclonal antibody, R10Z8E9 (Shankar et al., 2008; Lenz and Stubenrauch, 2011), generated at Genentech. Wash buffer (PBS/0.05% polysorbate 20, pH 7.4), assay diluent/sample diluent-1 (PBS/0.5% BSA/0.05% polysorbate 20/0.05% ProClin® 300, pH 7.4), and sample diluent-2 (PBS/0.5% BSA/0.05% polysorbate 20/0.05% ProClin® 300/0.25% CHAPS/0.35 M NaCl, pH 7.4) were prepared at Genentech.

Other reagents included horseradish peroxidase (HRP) conjugated recombinant Protein A/G (Thermo Fisher Scientific, Waltham, MA, USA); ChonBlock ELISA blocker & sample diluent (Chondrex, Inc., Woodinville, WA, USA); high sensitivity streptavidin (SA)-HRP (Pierce Biotechnology, Rockford, IL, USA); 3,3',5,5'-tetramethylbenzidine ([TMB] KPL Inc., Gaithersburg, MD, USA).

Thirty individual human sera from healthy donors and a pooled human serum were purchased from Bioreclamation IVT (Westbury, NY, USA). Pooled human serum from healthy individuals was used as the negative control (NC) for all assay formats. Positive controls (PC) from three sources were used: purified rabbit polyclonal antibody (pAb), a rabbit monoclonal antibody (mAb), and a fusion antibody of rabbit complementarity-determining regions (CDR) and human framework were all produced at Genentech.

2.2. Clinical study samples

Forty serum samples from a phase 2 trial of zinpentraxin alfa in subjects with PMF were obtained. In the Stage 1 trial, patients were treated with zinpentraxin alfa 10 mg/kg IV monthly with or without ruxolitinib (Tremblay and Mascarenhas, 2021). Serum samples for ADA assay evaluation were collected pre-dose on day 1 of cycle 1 (baseline), cycle 3, cycle 4, and on day 29 of cycle 6. The Stage 2 trial assessed zinpentraxin alfa in patients who were previously treated with or who were ineligible for ruxolitinib using three distinct dosing schedules (0.3 mg/kg, 3 mg/kg, and 10 mg/kg) (Tremblay and Mascarenhas, 2021).

Samples for evaluation in this study were collected at baseline and at post-treatment pre-dose on cycle 3 or cycle 4.

2.3. Step-wise bridging format (Fig. 1a)

Microtiter plates were incubated with 1.0 µg/mL of zinpentraxin alfa in phosphate-buffered saline (PBS) overnight at 4 °C. Plates were washed three times then blocked with assay diluent for one hour at room temperature (RT) with agitation. For samples with acid pre-treatment, serum was incubated in glycine (pH 2.6) for 15 min and then neutralized with a 1:1 ratio mixture of Tris (pH 9.0) and sample diluent-2 for 15 min. For samples without acid pre-treatment, serum was diluted 1:50 in sample diluent-2. Next, 100 µL of samples and controls were incubated for one hour at RT with agitation. Following plate washing, biotinylated zinpentraxin alfa diluted in assay diluent was added and incubated for one hour at RT. Streptavidin HRP was added to the plate after washing and the plate was then incubated at RT for one hour.

2.4. Direct binding format (Fig. 1b)

Microtiter plates were incubated with 1.0 µg/mL zinpentraxin alfa in PBS overnight at 4 °C. Plates were washed then blocked with 200 µL of ChonBlock or assay diluent for one hour at RT. Serum samples were diluted 1:50 in ChonBlock or acidified for 15 min then neutralized with a 1:1 ratio mixture of Tris (pH 9.0) and ChonBlock for 15 min. 100 µL of samples and controls were added and incubated for one hour at RT. Following plate washing, HRP conjugated Protein A/G was added and incubated for one hour at RT.

2.5. Total ADA format (Fig. 1c)

Monoclonal antibody to zinpentraxin alfa at 2.0 µg/mL in PBS was added to the microtiter plates and incubated overnight at 4 °C. Plates were washed and blocked with assay diluent for one hour at RT. Sample was diluted 1:50 in sample diluent-1. 100 µL of diluted samples were incubated for one hour at RT. Biotinylated R10Z8E9 was diluted, added to each well, and incubated for one hour at RT. Streptavidin HRP was added to the plate after washing and the plate was then incubated at RT for one hour.

2.6. Semi-homogeneous format (Fig. 1d)

Serum samples were diluted 25-fold in sample diluent-2. Biotinylated zinpentraxin alfa was diluted to 2.0 µg/mL in sample diluent-1. The diluted samples and biotinylated zinpentraxin alfa were mixed in a 1:1 ratio and incubated overnight at RT with agitation. The 96-well streptavidin plate was washed three times. 100 µL of sample/biotinylated zinpentraxin alfa mix was added and incubated at RT for one hour. Following plate washing, HRP conjugated Protein A/G was added and incubated for one hour at RT.

In each of the four assay formats, the final steps were identical. Colour development was enabled by adding the peroxidase substrate, TMB, and the enzymatic reaction was stopped using 1.0 M phosphoric acid. The signal was measured at 450nm using a SpectraMax plate reader (Molecular Devices, San Jose, CA, USA) to determine the absorbance, and the reference absorbance was measured at 630nm.

2.7. Screening assay cut point determination

To determine whether a sample is ADA-positive (i.e. has a signal at or above the cut point) or negative (i.e. has a signal below the cut point), it is necessary to establish an ADA screening assay cut point (CP) (Collet-Brose et al., 2016). Twenty baseline samples from a phase 2 study in PMF patients were each analyzed in duplicates. The average optical density (OD) value of individual samples was used in the data analysis. Eight replicates of the negative control, a pooled normal human serum,

were included on each plate.

To determine the assay cut point, normalized scores were obtained for each sample by calculating the logarithm of the ratio of the sample signal to the negative control signal. The standard deviation (SD) was used to estimate the within-individual variability. The CP value for each ADA assay format was determined to be the mean of the normalized scores from each individual serum plus 1.645 x SD, which represents a 5% non-treated positive rate (Shankar et al., 2008). The definition of the cut point factor (CPF) is as follows:

$$CPF = 10^{\text{mean of the normalized scores} + (1.645 \times SD)}$$

Cut point (CP) of each plate was defined as:

$CP = \bar{x} \times CPF$, $\bar{x}$ represents the mean value of the eight replicates of the negative control on each plate.

3. Results and discussion

The homogeneous bridging ELISA was evaluated first using biotinylated zinpentraxin alfa as the capture reagent and digoxigenin-labeled zinpentraxin alfa as the detection reagent; however, a low signal-to-noise ratio of positive controls due to high background signal made the assay unsuitable for detection of ADAs against zinpentraxin alfa (data not shown). Therefore, four alternative assay formats were developed and optimized as potential methods for ADA detection, including step-wise bridging, direct binding, total ADA, and semi-homogeneous formats. Details of each assay format are described in Fig. 1. Optimal assay conditions were established for each format before this evaluation, which included determination of optimal concentrations for the capture/detection reagents, the duration of incubation time for each step, and the minimum required dilution of sample (data not shown). In addition, key assay performance parameters to optimize the various assay formats are presented in this study including: 1) the utility of a blocking reagent in the direct binding assay format to reduce the non-specific background signal and to enhance the signal-to-noise ratio (Section 3.1); 2) the achievement of appropriate assay sensitivity and drug tolerance by evaluating the acid dissociation step in the step-wise bridging and direct binding formats (Section 3.2); 3) the evaluation of drug tolerance in the step-wise bridging, direct binding, and semi-homogeneous formats but not in the total ADA assay as in this format drug is added to the sample to form the drug/ADA immune complex (Section 3.3); and 4) the assessment of intra- and inter-assay precision for negative controls, and the evaluation of intra-assay precision for positive controls in each assay format (Section 3.4). In each section, assay conditions were first optimized as needed, followed by a comparison across different assay formats.

3.1. Direct binding format optimization

A direct binding assay typically involves the immobilization of the drug of interest (e.g. through passive absorption) on a solid surface, followed by its interaction with ADAs in the sample and detection using a detection antibody that is specific to the species of interest. While this format is convenient for high sample throughput (Li et al., 2011; Weeraratne et al., 2013), it is susceptible to generating high background signal and significant inter-subject variability which can lead to high assay cut point and low assay sensitivity. Moreover, a species-specific positive control and detection antibody are necessary, which restricts the detection of some ADA isotypes (e.g. IgG, IgM, IgA, etc.). Another limitation of the typical direct binding format is its inability to differentiate between ADAs that target the human antibody therapeutic and the therapeutic itself, as the anti-human detection antibody will bind to both resulting in false positive signals.

Johnson et al. described assays that optimized the direct binding format to detect ADAs against therapeutics that lack an immunoglobulin Fc domain (Johnson et al., 2021). This was achieved through the use of a

labeled Protein A/G reagent that could function as a single multi-species reagent. The authors verified that Protein A/G could be used as a versatile reagent to detect different isotypes of human immunoglobulins, with higher reactivity to IgG than IgM and IgA. A series of blocker diluents were evaluated for their ability to lower the non-specific background signal resulting in an increased signal-to-noise ratio of the specific response from ADA-positive samples. Among the evaluated blockers, ChonBlock was the most efficient reagent for reducing the background signal and maximizing the response ratio.

Our direct binding assay format is presented in Fig. 1b, in which zinpentraxin alfa was coated overnight and then blocked with a plate blocker (either a generic assay diluent or ChonBlock), followed by incubation with sample or assay controls. A rabbit polyclonal antibody (pAb) was used as a positive control during routine sample analysis. HRP-conjugated Protein A/G was next diluted and added to the plate. If a generic assay diluent (i.e. sample diluent-1 or sample diluent-2 from the "Materials and Methods") was used to block plates or to dilute samples, high background signal was observed due to the non-specific binding. The effectiveness of ChonBlock was evaluated at the plate blocking and sample dilution steps. As shown in Fig. 2a, a higher background signal from the negative control was observed if ChonBlock was only used at the plate-blocking step or as a sample diluent, while the background signal was significantly reduced if ChonBlock was included in both steps. Using ChonBlock did not impact the detection of the PC at 100 ng/mL and the signal-to-noise ratio was improved if ChonBlock was employed at both steps (Fig. 2b, c). These results confirmed the benefit of using ChonBlock to maximize the assay response ratio, as also observed by other groups (Terato et al., 2014; Waritani et al., 2017).

3.2. Evaluation of acid pre-treatment in step-wise bridging and direct binding assay formats to improve drug tolerance

Drug interference in the ADA assay can be more challenging when there are high levels of an endogenous drug counterpart which may interfere with the detection of ADAs (Zhong et al., 2017). The level of endogenous PTX-2 in human serum from healthy donors has been reported to be 20–50 µg/mL (Steel and Whitehead, 1994). While levels of endogenous PTX-2 are anticipated to be lower in patients with fibrotic conditions such as IPF, end-stage renal disease (Castano et al., 2009), and non-alcoholic steatohepatitis (Verna et al., 2015), they may still be considered high enough to introduce potential interference in the detection of ADAs.

During method development, acid dissociation was evaluated in the step-wise bridging (Fig. 1a) and the direct binding assay (Fig. 1b) formats. Assay responses from two positive controls, a monoclonal antibody, and a polyclonal antibody against zinpentraxin alfa, spiked in pooled normal human serum were compared with and without acid dissociation (Fig. 3a, b). Samples were either diluted 1:5 in acid solution, followed by a 1:10 dilution in neutralization buffer or 1:50 in sample diluent. In both assay formats, monoclonal PC showed equivalent signal with or without acid dissociation step. However, the polyclonal PC exhibited a significantly lower signal without acid dissociation when compared with acid pre-treated samples. These results suggest that the detection of polyclonal PC may be impacted by the presence of the endogenous PTX-2 and the acid pre-treatment dissociates the polyclonal PC bound to endogenous protein. This dissociation increases the accessibility of the capture and detection reagents to the ADAs.

Drug tolerance was evaluated at 40.0 µg/mL of zinpentraxin alfa, which is the estimated trough level based on the proposed study dosing regimen when the samples are collected for immunogenicity analysis (Verstovsek et al., 2023). The experiment was conducted using both polyclonal and monoclonal positive controls prepared at different concentrations in a pooled serum (containing approximately 34 µg/mL of endogenous PTX-2) and spiked with 40.0 µg/mL of zinpentraxin alfa, with or without an acid dissociation step (Fig. 4). Drug tolerance was determined to be the drug concentration where the assay signal fell

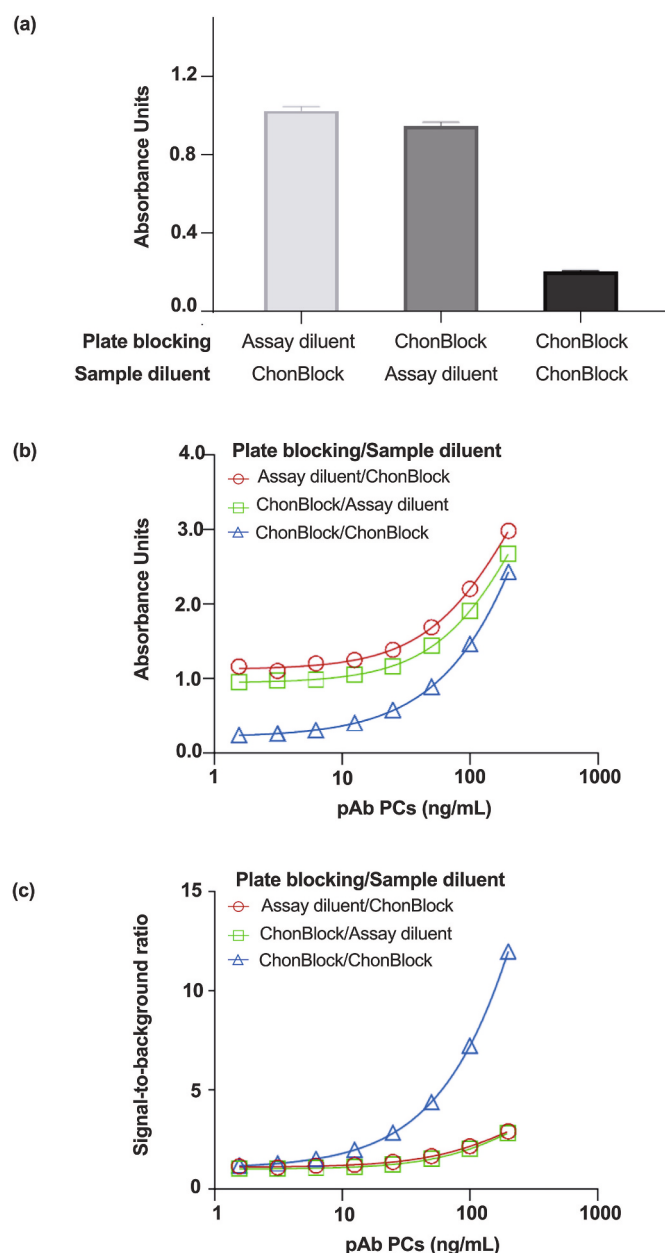


Fig. 2. Effectiveness of Chonblock used at plate blocking and sample dilution steps in the direct binding format. (a) Negative control signal under 3 assay conditions: ChonBlock used only as the sample diluent (light grey bar); ChonBlock used only at the plate blocking step (dark grey bar); or ChonBlock used as both sample diluent and plate blocker (black bar). Error bars represent the standard error of the mean ($N = 3$). Assay signal (b) and signal to background ratio (c) were determined using a pooled human serum spiked with purified polyclonal rabbit antibody against zinpentraxin alfa at 0–200 ng/mL under 3 assay conditions: ChonBlock used only as the sample diluent (red circles); ChonBlock used only at the plate blocking step (green squares); or ChonBlock used as both sample diluent and plate blocker (blue triangles). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

above or equal to the assay's specific cut point. In the step-wise bridging format, without acid pre-treatment, the assay signal was slightly above the cut point at the highest concentration of pAb PC (1000 ng/mL) but could not tolerate 40.0 $\mu\text{g/mL}$ of zinpentraxin alfa at lower concentrations, while mAb PC were able to tolerate 40.0 $\mu\text{g/mL}$ of zinpentraxin alfa at all tested concentrations (Fig. 4a). In the direct binding format, although a much lower signal was observed without acid pre-treatment,

the assay could tolerate 40.0 $\mu\text{g/mL}$ of zinpentraxin alfa at all tested concentrations of both pAb and mAb PCs (Fig. 4c). In both the step-wise bridging and direct binding formats, the drug tolerance was found to be improved by the acid pre-treatment step (Fig. 4b and d). As a result, the acid dissociation step was utilized for all subsequent evaluations of these two assay formats. Since the semi-homogeneous format showed a comparable signal between mAb PC and pAb PC and had appropriate drug tolerance at lower PC concentrations (25, 50, 100, and 200 ng/mL), acid dissociation was not evaluated for this format (Fig. 4e).

3.3. Evaluation of assay sensitivity

Surrogate animal-derived monoclonal or polyclonal antibodies are often used as positive control reagents in ADA assays to enable method development and validation. To assess the assay sensitivity of step-wise bridging, direct binding, and semi-homogeneous formats, a rabbit polyclonal antibody was used since polyclonal antibodies consist of a blend of antibodies that mimic the natural immune response to an antigen (Goyal et al., 2023). The evaluation of assay sensitivity was based on the lowest concentration of the surrogate control antibody that consistently yielded a positive signal above the assay cut point (Carrasco-Triguero et al., 2016). Relative sensitivity was 12.5 ng/mL, 25.0 ng/mL, and 100.0 ng/mL for the step-wise bridging format, direct binding format, and semi-homogeneous format, respectively (Table 1). In the total ADA assay format, R10Z8E9, a murine anti-human IgG1 monoclonal antibody was used as the detection reagent. The antibody targets a conformational epitope on the CH2 domain of all four subclasses of human Fc gamma (Stubenrauch et al., 2009). Thus, the positive control for this assay format was a chimeric antibody of the rabbit CDR with a human IgG framework. Table 1 showed that the relative sensitivity of the total ADA assay format was 125.0 ng/mL, which was higher than the sensitivity of the other three assay formats. However, the assay sensitivity comparison was difficult since a different positive control was employed for the total ADA assay format compared to the other three formats.

3.4. Intra- and inter-assay precision

Pooled human serum from healthy individuals was used as the negative control for all assay formats. For negative controls, intra-assay precision was evaluated by running the negative control eight replicates on each plate for each ADA assay format; inter-assay precision was analyzed from three runs performed on three different days. Intra-assay precision for the positive controls was evaluated at two different levels running in duplicates on each plate. For the step-wise bridging, direct binding, and semi-homogeneous assay formats, a rabbit polyclonal antibody was utilized as a positive control, whereas a chimeric antibody consisting of rabbit CDR and human framework was employed as a positive control for the total ADA format. Table 2 summarizes the results of intra- and inter-assay precision (%CV) of the negative control, as well as the intra-assay precision of positive controls. Overall, the intra-assay and inter-assay variability were <15% for the NCs and PCs for the direct binding, total ADA, and semi-homogeneous assay formats.

A higher degree of variability was observed from the negative controls in the step-wise bridging assay format with acid dissociation. The intra-assay %CV of the NC was over 20% in 11 out of 18 runs which were performed on different days. However, the variability from the positive controls were <15%, which was consistent with the other assay formats. Higher variability was also observed when 30 treatment naïve samples from MF patients were analyzed. Five out of 30 samples had higher than 20% CV from duplicate measurements. Different assay conditions, diluents, wash buffers, and assay blockers have been tested to troubleshoot the precision and robustness of this assay format, however, minimal improvement was achieved in assay variability. Although the reason for this phenomenon is not fully understood, we hypothesize that the presence of a variety ligands including auto-antigens, apoptotic cells,

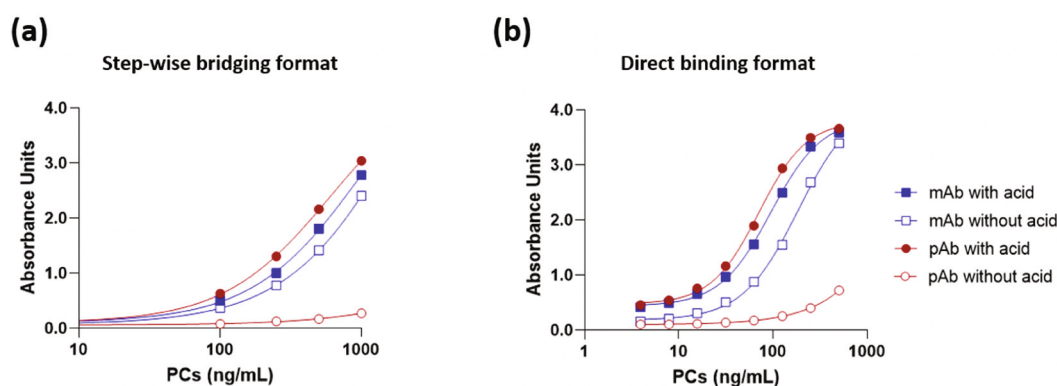


Fig. 3. Evaluation of two surrogate antibodies (mAb and pAb) for zinpentraxin alfa with and without acid dissociation in the step-wise bridging and direct binding assay formats. (a) Step-wise bridging format. (b) Direct binding format. A standard curve of a rabbit mAb (blue) or a rabbit pAb (red) was prepared with acid pre-treatment (solid) or without acid pre-treatment (open) before adding to the plates pre-coated with drug. The assay was performed following the procedure outlined in the “Materials and Methods” section. mAb = Monoclonal antibody positive controls; pAb = Polyclonal antibody positive controls. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

DNA, and micro-organisms of PTX-2 may interact with the capture and detection reagents and introduce higher-than-expected variability in this assay format (Steel and Whitehead, 1994; Alexandrescu, 2005; Garlanda et al., 2005).

3.5. Screening assay cut point factor determination

The cut point for the screening assay was determined for each assay format using serum collected from 20 MF patients at baseline and a pooled serum used as negative control for normalization. To determine the screening cut point factor, a false-positive rate of 5% was used to minimize the probability of obtaining false negative results (Smith et al., 2007; Gunn 3rd et al., 2016). Table 3 summarizes the cut-point factor for each assay format using 20 samples for this cut point evaluation due to the limited number of samples available during assay development. For the selected ADA format assay validation, a higher number of individuals will be included for screening assay cut point factor determination.

The signal from each sample was measured in duplicate for each assay format (Fig. 5). Although signal from the negative control was consistent across the four assay formats, a higher inter-donor variability was observed in the semi-homogeneous format (Fig. 5d) compared to the other three formats (Fig. 5a-c). It is possible that the absence of an effective blocker in the semi-homogeneous format led to increased non-specific binding, potentially due to variations in immunoglobulin concentrations or the presence of immune complexes that could bind to the plates and subsequently interact with Protein A/G. As a result, a higher assay cut point factor was derived in the semi-homogeneous format compared to the other three formats.

3.6. Application of different ADA assay formats for clinical sample analysis

Baseline and post-treatment samples from 20 MF patients in a phase 2 study of zinpentraxin alfa were selected and analyzed using different assay formats to further evaluate and compare their performance. Of the post-treatment samples, 10 were negative, 10 were screened and confirmed positive, and ADA titer was further characterized using the SPEAD assay previously developed by Promedior and used to evaluate immunogenicity in the phase 2 study (Table 4).

All ADA negative samples also tested negative using the four ADA assay formats, except for a sample from one patient in the step-wise bridging format. Interestingly, different numbers of ADA-positive samples were detected with various assay formats. Three out of the 10 ADA-positive samples tested positive using the step-wise bridging format,

however, the signal was overall low and slightly above the plate cut point (AU = 0.171). Using the direct binding format, only two samples tested positive with acid pre-treatment, compared with 5 samples without acid pre-treatment. This result was inconsistent with what we observed for the polyclonal positive controls, in which the polyclonal positive controls showed optimal signal with acid pre-treatment. These data indicate that the acid dissociation step might have different impact on the positive controls compared to ADAs from study samples. This is one of the limitations of including such sample pre-treatment step in ADA assays. The total ADA and semi-homogeneous formats showed the best overall agreement with the SPEAD assay results, detecting 6 and 7 out of 10 ADA-positive samples, respectively. Three samples with the lowest titer in the SPEAD assay tested negative with the semi-homogeneous assay format.

Variability in immunogenicity data obtained from ADA assays is often observed due to differences in assay formats, assay conditions (such as acid pre-treatment), or screening cut points. In this study, based on the assay format, the types of ADAs measured could be different. In the total ADA format, free ADAs and drug-bound ADAs, ADAs bound to the endogenous PTX-2 can all be detected; however, this format is capable of detecting only ADAs of the IgG isotype. Both the step-wise bridging and direct binding formats are referred to as solid-phase assays, in which zinpentraxin alfa was immobilized on a solid surface. The total ADA and semi-homogeneous formats allow for binding between the drug and ADAs in solution. The solid-phase assay formats could potentially mask the ADA epitopes and interfere with the binding between zinpentraxin alfa and ADAs in the samples, which could potentially lead to a lower incidence of ADA detection. The total ADA assay and semi-homogeneous formats allow the binding between ADAs and capture reagents to take place in the solution phase, which circumvents the epitope masking problems and results in more ADA-positive samples being detected.

4. Conclusions

As therapeutic modalities become more diverse and structurally complex, the bridging immunoassay may not always be appropriate for ADA measurement, despite its wide use. Therefore, it is necessary to continue developing new methods for the detection of ADAs with different assay designs and technology platforms to support immunogenicity assessments in clinical development. In this study, the homogeneous bridging assay format was initially evaluated, but it was deemed inappropriate due to higher-than-expected background and low signal-to-noise ratio observed with this assay format.

This paper describes four assay formats: step-wise bridging, direct

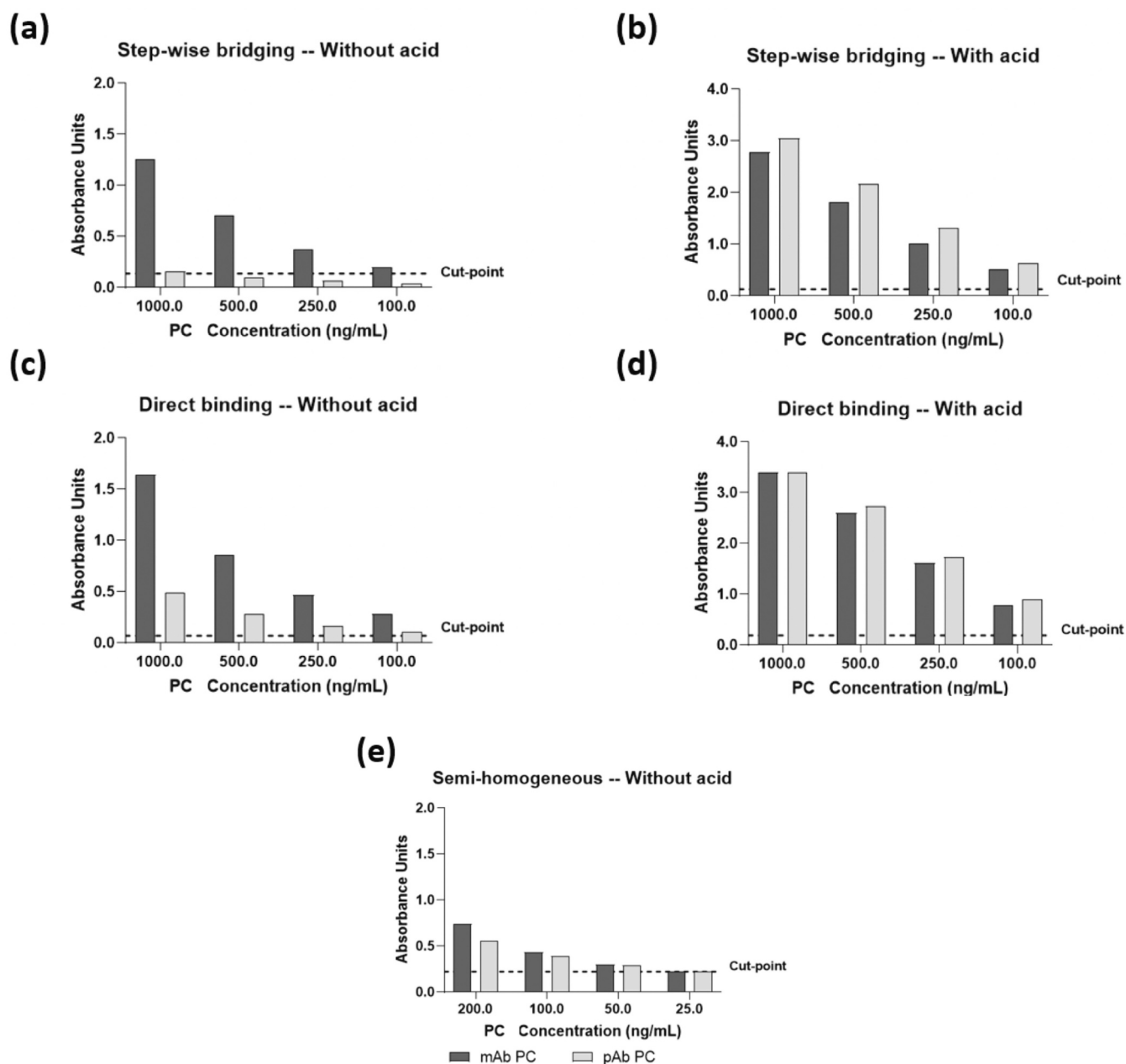


Fig. 4. Drug tolerance was evaluated at 40 $\mu\text{g/mL}$ zinpentraxin alfa at different levels of monoclonal PC and polyclonal PC in the step-wise bridging (a, b) and direct binding formats (c, d) with and without acid dissociation, and semi-homogeneous format (e) without acid dissociation. Dotted lines represent plate cut point. mAb PC = Monoclonal antibody positive control; pAb PC = Polyclonal antibody positive control.

Table 1
Relative sensitivity of various ADA assay formats.

Step-wise bridging			Direct binding			Total ADA			Semi-homogenous		
pAB PC (ng/mL)	OD450	S/N	pAB PC (ng/mL)	OD450	S/N	Chemeric PC (ng/mL)	OD450	S/N	pAB PC (ng/mL)	OD450	S/N
200.0	1.129	8.2	200.0	0.988	4.8	1000.0	0.570	4.2	400.0	1.269	4.8
100.0	0.574	4.2	100.0	0.682	3.3	500.0	0.400	2.9	200.0	0.799	3.3
50.0	0.333	2.4	50.0	0.419	2.0	250.0	0.266	1.9	100.0	0.480	2.0
25.0	0.277	2.0	25.0	0.331	1.6	125.0	0.206	1.5	50.0	0.318	1.6
12.5	0.211	1.5	12.5	0.217	1.1	62.5	0.177	1.3	25.0	0.241	1.1
6.3	1.63	1.2	6.3	0.235	1.1	31.3	0.159	1.2	12.5	0.200	1.1
3.1	1.29	0.9	3.1	0.231	1.1	15.6	0.142	1.0	6.3	0.181	1.1
1.6	1.7	1.2	1.6	0.212	1.0	7.8	0.141	1.0	3.1	0.172	1.0
Cut point: 0.190			Cut point: 0.245			Cut point: 0.190			Cut point: 0.374		

Sensitivity of the various assay formats is determined based on the lowest concentration of the positive control (PC) exhibiting a response above the plate cut point, and the values are indicated in **bold** and underlined.

Table 2

Summary of intra-assay and inter-assay analysis of negative and positive controls in different assay formats.

Step-wise bridging			Direct binding			Total ADA			Semi-homogenous		
(ng/mL)	Intra-assay (%CV)	Inter-assay (%CV)	(ng/mL)	Intra-assay (%CV)	Inter-assay (%CV)	(ng/mL)	Intra-assay (%CV)	Inter-assay (%CV)	(ng/mL)	Intra-assay (%CV)	Inter-assay (%CV)
NC	≤ 81.0	29.0	NC	≤ 10.2	11.2	NC	≤ 4.9	6.8	NC	≤ 5.2	5.0
PC_250	≤ 12.0	ND	PC_200	≤ 11.1	ND	PC_1000	≤ 4.8	ND	PC_400	≤ 6.0	ND
PC_31.2	≤ 9.2	ND	PC_50	≤ 7.1	ND	PC_250	≤ 8.5	ND	PC_65	≤ 9.0	ND

The intra-assay precision of NCs was calculated from eight replicates included on each plate; the intra-assay precision of PCs was calculated at two different levels assayed in duplicates. The inter-assay precision of NCs were analyzed as the %CV of the signal observed in three independent runs. NC = Negative controls; PC = Positive controls; ND = Not determined.

Table 3

Cut point factor in MF patients using different ADA assay formats.

Assay format	%CV across baseline samples	Cut point factor
Step-wise bridging	16%	1.24
Direct binding	41%	1.19
Total ADA assay	49%	1.39
Semi-homogenous	71%	2.08

Determination of inter-donor variability and cut point factor based on 5% false positive rates from 20 MF patients at baseline across different ADA assay formats.

binding, total ADA, and semi-homogeneous formats, which were developed and evaluated to detect ADAs induced by zinpentraxin alfa. Key results from the evaluation of each assay format are presented in Table 5. While there are some similarities among the four assay formats when compared, each format also possesses unique properties that set it apart from the others. The direct binding format requires a special blocker at the plate blocking and sample dilution steps to reduce the background signal and enhance signal-to-noise ratio, while the other three formats use more generic buffers in the assay protocol. The acid dissociation step is essential to improve the signal of a polyclonal

surrogate antibody in both the step-wise bridging and direct binding formats. Although the step-wise bridging format has the most similarity with the homogenous bridging format, the poor precision observed from negative control in this format could result in higher and variable assay cut points and potential false-negative results in the screening assay. In addition, similar to the homogenous bridging assay, the step-wise bridging format does not detect ADA from the IgG4 subclass, which could be induced by repeated administration of a highly immunogenic therapeutics (Aalberse et al., 1983).

The sensitivity of the total ADA format was slightly higher than the sensitivity requirements set by FDA for clinical assays (≤ 100 ng/mL). Drug tolerance at 40 μ g/mL of zinpentraxin alfa could be achieved at the tested concentrations of both monoclonal and polyclonal antibody controls in the step-wise bridging format with acid pre-treatment, direct binding format with and without acid pre-treatment, and semi-homogeneous format. Based on the considerations of assay sensitivity, drug tolerance, precision, and ability to detect various ADA isotypes, we endorsed the semi-homogenous assay over the other three assay formats, which was the most conservative and identified more ADA-positive samples when analyzing a subset of MF patient samples from a phase 2 study of zinpentraxin alfa. A multi-tiered ADA testing approach will be used in the immunogenicity analysis of clinical samples, where samples

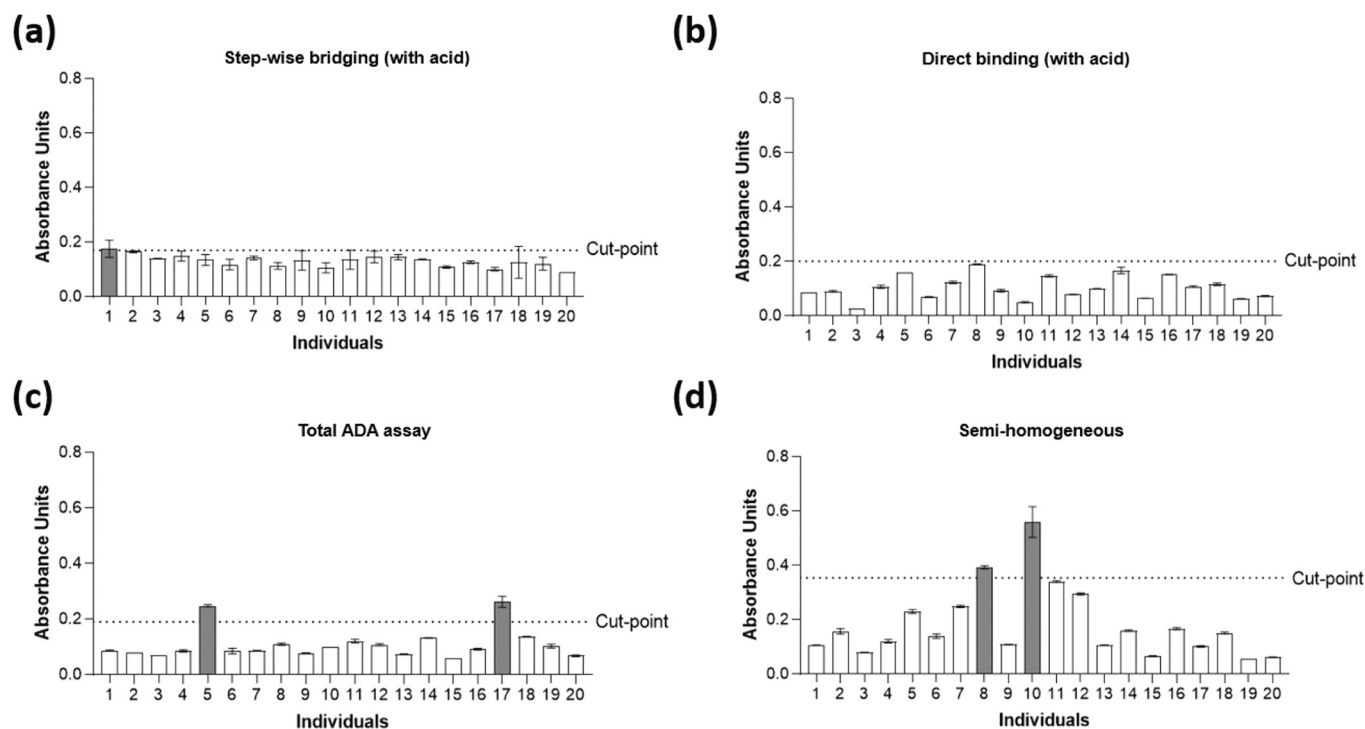


Fig. 5. Assay signal obtained from 20 treatment naïve MF samples using different ADA assay formats. (a) Step-wise bridging with acid pre-treatment; (b) Direct binding with acid pre-treatment; (c) Total ADA assay; (d) Semi-homogeneous format. Individual serum samples that produced a signal above the plate cut point, which is represented by the dotted line, were deemed positive, and the corresponding bars were shaded in grey in the graphs.

Table 4

Comparison of ADA results in 20 clinical samples using different ADA assay formats.

Subject	TP	SPEAD		Step-wise bridging		Direct binding				Total ADA		Semi-homogenous	
		Titer	Screening Result (+/−)	AU	Screening Result (+/−) Plate CP = 0.171	With acid		Without acid		AU	Screening Result (+/−) Plate CP = 0.190	AU	Screening Result (+/−) Plate CP = 0.339
						AU	Screening Result (+/−) Plate CP = 0.201	AU	Screening Result (+/−) Plate CP = 0.119				
1	C4D1	3.8	+	0.149	−	0.142	−	0.306	+	1.349	+	2.227	+
2	C4D1	3.1	+	0.186	+	0.105	−	0.104	−	0.767	+	0.474	+
3	C4D1	3.1	+	0.151	−	1.329	+	3.234	+	0.592	+	3.205	+
4	C3D1	2.4	+	0.104	−	0.201	+	0.503	+	0.851	+	0.954	+
5	C6D29	2.4	+	0.171	+	0.051	−	0.077	−	1.153	+	0.775	+
6	C4D1	1.7	+	0.168	−	0.139	−	0.074	−	0.540	+	0.348	+
7	C4D1	1.7	+	0.102	−	0.059	−	0.075	−	1.115	−	0.650	+
8	C4D1	1.7	+	0.130	−	0.146	−	0.051	−	0.095	−	0.253	−
9	C3D1	1.0	+	0.181	+	0.155	−	0.216	+	0.170	−	0.195	−
10	C4D1	1.0	+	0.137	−	0.138	−	0.126	+	0.155	−	0.207	−
11	C4D1			0.169	−	0.153	−	0.054	−	0.124	−	0.288	−
12	C4D1			0.147	−	0.078	−	0.050	−	0.102	−	0.140	−
13	C4D1			0.161	−	0.083	−	0.041	−	0.073	−	0.081	−
14	C4D1			0.153	−	0.155	−	0.033	−	0.138	−	0.163	−
15	C4D1			0.165	−	0.077	−	0.099	−	0.070	−	0.065	−
16	C4D1			0.189	+	0.151	−	0.046	−	0.176	−	0.207	−
17	C4D1			0.150	−	0.100	−	0.043	−	0.145	−	0.110	−
18	C4D1			0.096	−	0.108	−	0.034	−	0.141	−	0.145	−
19	C4D1			0.129	−	0.074	−	0.017	−	0.133	−	0.057	−
20	C4D1			0.110	−	0.074	−	0.017	−	0.079	−	0.090	−

ADA results were obtained from the four ADA assay formats: step-wise bridging, direct binding with and without acid pre-treatment, total ADA assay, and semi-homogeneous in the serum from patients treated with zinpentraxin alfa. 10 samples were screened and confirmed positive by the SPEAD assay, with titer results characterized; 10 samples were screened negative by the SPEAD assay. Sample was considered ADA positive (+), if the signal is at or above the plate cut point; or negative (−) if the signal is below the plate cut point. AU = absorbance units; CP = cut point; TP = time points. AU of samples tested positive were shown in **Bold**.

Table 5

Summary of key parameters for evaluating the four assay formats of anti-zinpentraxin alfa antibodies detection.

Evaluation parameters	Step-wise bridging	Direct binding	Total ADA	Semi-homogenous
ADA Isotypes to be detected	All, except IgG4	All, except IgE	IgG	All, except IgE
Acid dissociation	Yes	Yes	No	No
Special blocker (plate blocking and sample dilution)	No	ChonBlock	No	No
Assay sensitivity ¹	12.5 ng/mL	25.0 ng/mL	125.0 ng/mL	100.0 ng/mL
Drug tolerance	40.0 µg/mL	40.0 µg/mL	Not evaluated	40.0 µg/mL
Intra-assay precision of negative controls (%CV)	≤ 81.0%	≤ 10.2%	≤ 4.9%	≤ 5.2%
Inter-assay precision of negative controls (%CV)	29.0%	11.2%	6.8%	5.0%
Intra-assay precision of positive controls (%CV)	≤ 12.0%	≤ 11.1%	≤ 8.5%	≤ 9.0%
Screening cut point factor	1.24	1.19	1.39	2.08

¹ Rabbit polyclonal antibody was used in the Step-wise bridging, direct binding and semi-homogeneous assay formats; a fusion antibody of rabbit CDR and human framework was used in the total ADA format.

that test positive in the screening assay will undergo a confirmatory assay to demonstrate the specificity of ADAs for the therapeutic (2019).

In conclusion, after systematically comparing four different ADA assay formats, the semi-homogeneous assay format was selected for immunogenicity analysis of zinpentraxin alfa. The semi-homogeneous assay format is sensitive, reproducible, and provides good drug tolerance, without the use of acid pre-treatment or a special blocker. Although the other three assay formats did not provide appropriate

performance for zinpentraxin alfa, these formats may provide additional options for immunogenicity assessment and offer a more streamlined methodology to expedite assay development for different therapeutic modalities.

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CRediT authorship contribution statement

Xiaoyun Yang: Conceptualization, Investigation, Data curation, Formal analysis, Writing – original draft, Writing – review & editing. **Ketevan Siradze:** Investigation, Data curation, Formal analysis. **Gizette Sperinde:** Conceptualization. **Audrey Arjomandi:** Resources, Writing – review & editing. **Saloumeh Fischer:** Supervision, Conceptualization, Project administration, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors are employed by Genentech, Inc., and own stock in F. Hoffmann-La Roche Ltd. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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